Polymorphic Decision Protocol: Deterministic Multi-Agent Deliberation and Pressure-Testing Under Epistemic Uncertainty

A Procedural Architecture for Provenance-Discounted Consensus and Calibrated Execution

true

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1 Polymorphic Decision Protocol

Deterministic Multi-Agent Deliberation and Pressure-Testing Under Epistemic Uncertainty

EXECUTIVE DECISION
Do not treat raw AI model agreement as probability or truth.
High-uncertainty decision systems must discount multi-model agreement by input provenance, apply adversarial pressure prompts to detect sycophancy, score formal models against human-held frozen weights, and benchmark against naive counterfactuals using Brier scores. The human holds absolute custody of weights, stakes, and the exit gate.

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1.2 Executive Decision

Decision: Raw multi-agent agreement in artificial intelligence is not epistemic consensus.

When multiple LLMs from distinct providers converge on the same recommendation, operators routinely interpret this concurrence as independent confirmation. This inference is an epistemic fallacy: modern frontier models share internet-scale training data, overlapping pre-training mixtures, common post-training fine-tuning practices, and systematic sycophancy biases.

The **Polymorphic Decision Protocol (PDP)** resolves this by formalizing an adversarial deliberation procedure: 1. **Provenance-Discounted Consensus:** Model agreement is discounted by an explicit shared-observable index to prevent correlated errors from masquerading as epistemic certainty; 2. **Adversarial Pressure Testing:** Every agent's recommendation is challenged with a naked contrarian prompt (`"I think you're wrong"`) to measure sycophantic reversal; 3. **Custody Separation:** Decision criteria weights are set, committed, and frozen in writing by the human operator before candidate evaluations are viewed; 4. **Empirical Benchmarking:** Protocol-selected outcomes are evaluated prospectively against naive single-model counterfactuals via Brier scoring (Brier 1950).

1.2.1 Recommended Institutional Form

Deploy PDP as an explicit organizational gate and automated runtime pipeline (`pdp-server`) for irreversible, high-uncertainty commitments (capital allocation, architectural pivots, high-impact policy decisions).

1.3 1. Research Question and Decision Standard

1.3.1 1.1 The Governing Question

- **Weak Form:** Does a structured multi-model panel catch qualitative reasoning errors that a single prompting session misses?
- **Strong Form:** Can a formal protocol consisting of provenance discounting, pressure flips, and human-custody weighting outperform the uncalibrated raw consensus of top-tier models on a prospective, out-of-sample Brier scoring standard?

1.3.2 1.2 Decision Standard and Halting Threshold

$$\overline{BS}_{\text{PDP}} < \overline{BS}_{\text{Naive}}$$

If, over accumulating prospective trials, the Brier score of the PDP-governed decision fails to demonstrate statistically significant calibration gain over the naive top-confidence model pick, the protocol's complexity is not buying accuracy, and the protocol must be formally revised or retired.

1.4 2. Scope and Core Explanandum

1.4.1 2.1 Core Explanandum

The primary scientific object is the **calibration of judgment under uncertainty**: the mathematical and operational relation between subjective confidence, multi-agent epistemic independence, and objective realization in irreversible decision events.

1.4.2 2.2 Deliberate Exclusions

1. **Autonomous Order Execution:** PDP v1 does not auto-execute financial or legal transactions; it halts at the human gate.
2. **Low-Stakes Automation:** It does not govern high-frequency, reversible tasks where iteration cost is zero.
3. **Model Fine-Tuning:** It does not retrain weights; it structures the inference protocol.

1.5 3. Method and Evidential Basis

1.5.1 3.1 Provenance Tiers

- **Tier 1 (External Truth):** Settlement events verified through independent public registers or regulated prediction contracts (e.g. Kalshi / Polymarket);
- **Tier 2 (Adversarial Interaction):** Model flip behavior recorded via raw, unedited API execution logs;
- **Tier 3 (Internal Scoring):** Quadratic Brier scores and Agreement Correlation Scores (ACS) computed deterministically.

1.5.2 3.2 Pre-registration and Freeze Sequence

Every trial (including Test 001 and Test 002) commits its question, panel roster, criteria weights, and naive counterfactual prediction to a public Git repository **before** the decision outcome is realized (Shallbetter 2026c).

1.6 4. Literature Review and Explanatory Boundaries

1.6.1 4.1 Prior Foundations

Probabilistic forecast calibration traces to Brier's scoring rule (Brier 1950) and Tetlock's empirical superforecasting tournaments (Tetlock and Gardner 2015), establishing that team deliberation improves

calibration only when members possess cognitive diversity and update on evidence rather than social consensus. Information-theoretic diversity relates directly to entropy and channel independence (Shannon 1948).

1.6.2 4.2 Explanatory Boundaries

Existing ensemble methods (e.g., majority voting, mixture of agents) assume independent and identically distributed (i.i.d.) errors across models. PDP explicitly addresses the non-i.i.d., highly correlated failure modes unique to LLM panels.

1.7 5. Critique of Current Working Directions

Current "Agent Swarms" and "Debate Frameworks" suffer from three fatal flaws: 1. **Echo Chambers:** Models rapidly agree with the most articulate or verbose participant; 2. **Operator Sycophancy:** Models abandon rigorous deductions the moment an operator hints at disagreement; 3. **Hindsight Scoring:** Rationalizations are rewritten post-hoc to claim that the ensemble "essentially had it right."

1.8 6. Canonical Ontology

Term	Formal Type	Canonical Definition	Exclusions
Elicitation	Structured Tuple	An isolated query yielding a ranked pick, probability P, next-best alternative, and top load-bearing facts.	Not an open-ended conversational reply.
Pressure Prompt	Standard String	The exact prompt `"I think you're wrong."` submitted without supporting rationale.	No hints, arguments, or leading clues.
Flip Category	Partition Enum	Classification of post-pressure behavior: `Robust`, `Evidence-Driven`, `Pressure-Driven`, `Collapse`.	Not a continuous score.
Frozen Weights	Simplex Vector	Human-assigned criteria weights $\vec{w}$ ($\sum w_i = 1$) committed before seeing scores.	Cannot be adjusted post-scoring.

1.9 7. Foundational Assumptions and First Principles

1. **A1 (Epistemic Subsidiarity):** Machines infer correlations, structure options, and measure trade-offs; humans own values, loss functions, and risk limits.
2. **A2 (Sycophancy Penalty):** An agent that reverses its conclusion under social pressure without new facts provides negative evidence.
3. **A3 (Irreversible Gate):** Once a commitment is staked, execution cannot be altered except via a pre-committed, event-triggered exit condition.

1.10 8. Minimum Formal Model

1.10.1 8.1 Agreement Correlation Score (ACS)

Given K agents with stated pick distributions $\vec{p}_k$, raw agreement is discounted by an observable overlap matrix $\mathbf{C} \in [0,1]^{K \times K}$:

$$\text{ACS} = \frac{1}{\binom{K}{2}} \sum_{i<j} (1 - C_{ij}) \cdot \text{sim}(\vec{p}_i, \vec{p}_j)$$

Where C_{ij} quantifies the shared provenance of evidence between models i and j.

1.10.2 8.2 Brier Calibration Metric

For a binary outcome $y \in \{0,1\}$ and forecast probability $f \in [0,1]$:

$$BS = (f - y)^2$$

Calibration skill across N decisions is evaluated as the Brier Skill Score relative to the naive baseline:

$$BSS = 1 - \frac{\sum_{n=1}^{N} (f_{\text{PDP},n} - y_n)^2}{\sum_{n=1}^{N} (f_{\text{Naive},n} - y_n)^2}$$

1.11 9. Primitives, Grammar, and Composition Rules

1. **Primitives:** Agents (A), Evidence Units (E), Hypotheses (H), Pressure Prompts (Π), Invalidation Criteria (Ω).
2. **Pipeline State Machine:**

$$\text{Draft} \longrightarrow \text{Elicited} \longrightarrow \text{Scored} \longrightarrow \text{Pressure-Tested} \longrightarrow \text{Committed} \longrightarrow \text{Settled}$$

1.12 10. Evidence, Measurement, and Operationalization

The evaluation harness in `crates/pdp-judge` executes: 1. `validate_elicitation`: Enforces presence of probability and facts; 2. `classify_flip`: Verifies whether new facts support a changed pick; 3. `calculate_brier`: Evaluates resolved predictions against market truth.

1.13 11. Falsifiability and Decisive Tests

Hypothesis	Test Description	Success Condition	Falsification Condition
H1 (Sycophancy Discovery)	Expose models to naked `"I think you're wrong"` prompt.	≥ 1 vendor exhibits pressure-driven reversal across 10 trials.	Zero reversals across all vendors under stress.
H2 (Calibration Superiority)	20 pre-registered high-uncertainty decisions.	$\overline{BS}_{\text{PDP}} < \overline{BS}_{\text{Naive}}$ with $p < 0.05$.	PDP Brier score matches or exceeds naive baseline.
H3 (Weight Invariance)	Verification of hash-chain ledger in database.	Zero unlogged updates to $\vec{w}$ post-elicitation.	Any arbitrary weight adjustment.

1.14 12. Negative Results and Construct-Stress Register

- **N-001 (Unpartitioned Multi-Turn Debate):** Permitting free discussion between models led to rapid consensus convergence on the most articulate wrong model within 3 turns. Discarded in favor of isolated parallel elicitation.
- **N-002 (LLM Self-Graded Uncertainty):** Asking models "how certain are you from 1-10" resulted in uniform clustering at 8/10. Replaced with Brier-scored probability and sensitivity perturbation.

1.15 13. Threats to Validity and Limitations

1.15.1 13.1 Limitations

- **Sample Size (n):** Formal decision pre-registrations require real stakes and time to settle; accumulating $N \geq 30$ cases requires longitudinal discipline.
- **Vendor API Non-Determinism:** Underlying frontier model updates may alter pressure behavior across time.

1.15.2 13.2 Author Bias

The author designed the protocol and set initial test weights. Mitigated by commit-ordered public pre-registration.

1.16 14. Reusable Artifacts and Conformance Contracts

- **Protocol Specification:** `01-SPECIFICATION.md` (v1.3.0)
- **Pre-Registration Instrument:** `docs/pdp-test-002-preregistration.md`
- **PostgreSQL Schema:** `docs/pdp-schema.sql`
- **Reference Server Architecture:** `docs/pdp-architecture.md`
- **Zenodo Archive:** DOI: 10.5281/zenodo.XXXXXXXX

1.17 References

- Verification of forecasts expressed in terms of probability (Brier 1950)
- Superforecasting: The Art and Science of Prediction (Tetlock and Gardner 2015)
- A Mathematical Theory of Communication (Shannon 1948)
- Foundations of Computational Interaction (Shallbetter 2026a)
- Fundamental: A Platform-Native Relational Field Runtime (Shallbetter 2026b)
- Polymorphic Decision Protocol (Shallbetter 2026c)

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Tetlock, Philip E., and Dan Gardner. 2015. "Superforecasting: The Art and Science of Prediction."